

Integrating artificial intelligence and gamification in rehabilitation: A scoping review

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ABSTRACT

The integration of artificial intelligence (AI) and gamification represents a promising direction in rehabilitation, enabling personalized training, real-time feedback, and enhanced patient engagement. However, research in this area remains fragmented, necessitating a structured synthesis. This scoping review analyzed 20 studies published over the past two decades and indexed in PubMed, Cochrane Library, Web of Science, and IEEE Xplore. Stroke rehabilitation was the most frequently addressed condition (9 studies), followed by motor dysfunction (2), rheumatoid arthritis (2), cognitive impairment (3), phantom limb pain (2), ADHD (1), and multiple sclerosis (1). AI techniques included machine learning (10 studies), deep learning (4), and unspecified methods (6), while game-based interventions were categorized as gamification (4), serious games (8), and digital games (8). Reported outcomes indicated improvements in motivation, therapy adherence, and various metrics of rehabilitation performance. Nonetheless, key challenges persist, including small sample sizes, methodological heterogeneity, data privacy concerns, and the lack of large-scale clinical trials. Barriers related to cost, accessibility, and ethics also hinder broader implementation. Future research should prioritize multimodal data integration, AI-driven remote rehabilitation platforms, and wearable technologies to enhance scalability and clinical impact.

1. Introduction

Rehabilitation plays a pivotal role in modern healthcare, aiming to restore functional abilities and enhance the quality of life for individuals with disabilities or impairments. It encompasses a wide range of interventions—including physical therapy, occupational therapy, cognitive rehabilitation, and speech-language training—targeted at improving motor, sensory, cognitive, or emotional function [1]. Rehabilitation may occur in hospital settings, outpatient clinics, or even at home, and typically requires sustained engagement and clinician supervision to achieve long-term outcomes [2]. However, traditional rehabilitation approaches face several persistent challenges, including high costs, limited accessibility, and patient disengagement resulting from repetitive and monotonous exercises [3]. These barriers often lead to suboptimal therapy adherence and hinder long-term rehabilitation outcomes. In response to these challenges, game-based rehabilitation has emerged as a promising alternative, integrating interactive and motivational elements to enhance patient engagement and improve adherence to rehabilitation programs [4,5].

Gamification in rehabilitation has gained increasing attention due to its ability to foster motivation and engagement. Games introduce reward mechanisms, goal-setting strategies, and interactive feedback, which can transform rehabilitation exercises into a more enjoyable and stimulating experience [6]. Studies have demonstrated that digital and gamified rehabilitation interventions show promise in enhancing motor function, cognitive abilities, and overall therapy outcomes in patients with stroke, motor impairments, and neurodegenerative diseases [7]. For instance, stroke patients who participate in game-based therapy have exhibited substantial improvements in motor function and neuroplasticity, highlighting the therapeutic potential of gamification in rehabilitation [8].

Simultaneously, artificial intelligence (AI) is transforming various aspects of healthcare, including disease diagnosis, personalized treatment planning, and patient monitoring [9,10]. In the field of rehabilitation, AI has shown tremendous potential in optimizing therapy protocols, delivering real-time feedback, and enabling adaptive rehabilitation exercises [11,12]. AI-driven rehabilitation systems are capable of analyzing patient performance data, dynamically adjusting exercise

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difficulty, and generating personalized interventions, thereby improving both the effectiveness and efficiency of rehabilitation programs [13]. Additionally, AI-powered assessment tools enhance the accuracy and objectivity of patient evaluations, enabling evidence-based decision-making in clinical rehabilitation settings [14,15]. Moreover, the integration of AI in rehabilitation has been suggested to enhance patient engagement and daily activity participation, reinforcing therapy adherence [16,17].

Recent advancements in AI have further shaping the landscape of game-based rehabilitation, leading to the development of intelligent, adaptive, and immersive rehabilitation environments. AI is increasingly being integrated into gaming applications to generate dynamic game scenarios, personalize user experiences, and automate performance assessments [18]. For example, AI-controlled non-player characters (NPCs) can adapt to user behavior in real-time, offering interactive and realistic rehabilitation scenarios that enhance immersion and motivation [19]. Furthermore, AI-driven gamification strategies dynamically adjust game mechanics based on real-time patient responses, fostering continuous engagement and personalized therapy experiences [20].

The integration of AI and gamification in rehabilitation is further supported by several well-established theoretical frameworks. For instance, Self-Determination Theory posits that fulfilling basic psychological needs—namely autonomy, competence, and relatedness—can intrinsically motivate individuals to engage in behaviors such as therapy adherence [21]. Game elements that incorporate rewards, challenges, and social interaction can activate these motivational drivers. Similarly, Flow Theory emphasizes the importance of achieving a balance between challenge and skill, suggesting that personalized difficulty levels and immediate feedback—both of which can be dynamically adjusted through AI—can foster an immersive and optimally engaging rehabilitation experience [22]. From a clinical perspective, Motor Learning Theory emphasizes the importance of repetitive practice, augmented feedback, and task-specific training in promoting motor skill acquisition and retention [23]. These principles provide theoretical support for the use of interactive, feedback-rich environments—such as AI-enhanced game-based systems—in rehabilitation contexts aimed at improving motor performance.

Despite the promising potential of integrating AI and gamification in rehabilitation, research in this area remains relatively fragmented. While extensive studies have examined AI-driven rehabilitation systems and gamification techniques separately, the combined application of these two technologies in rehabilitation has received limited systematic investigation. Given the growing emphasis on personalized, engaging, and accessible rehabilitation approaches, further exploration is warranted to assess how AI-powered gamification can enhance rehabilitation practices and improve patient outcomes.

1.1. Contribution of this scoping review

With the increasing integration of AI and gamification in rehabilitation, a systematic synthesis of existing research is crucial to understanding current applications and identifying areas requiring further exploration. While AI-driven rehabilitation systems and gamification strategies have been widely studied, the synergy between these two technologies in rehabilitation remains underexplored. AI has been applied in rehabilitation to analyze patient data, provide real-time feedback, and personalize therapy sessions, while gamification has been shown to enhance patient engagement, motivation, and therapy adherence. However, limited research has systematically examined how AI can enhance gamification mechanisms in rehabilitation settings, or how their integration impacts rehabilitation effectiveness.

To bridge this gap, this scoping review aims to systematically examine existing literature and address the following research questions:

RQ1: What are the current applications of AI-driven gamification in rehabilitation, and how are these technologies being integrated?

RQ2: What benefits and challenges have been reported in studies on AI-powered gamification in rehabilitation?

RQ3: What gaps exist in the current research, and what are the potential directions for future exploration?

2. Methods

A scoping review is an effective approach for mapping the breadth and scope of literature on a given topic. It provides a clear overview of the volume of existing research while identifying key themes and trends [24]. Due to its exploratory nature, a scoping review is particularly useful for summarizing current knowledge and highlighting gaps that warrant further investigation. We conducted this scoping review in accordance with the PRISMA Extension for Scoping Reviews (PRISMA-ScR) reporting guidelines and checklist to ensure transparency and methodological rigor [25]. This review aims to systematically map the latest advancements in the integration of artificial intelligence and gamification in rehabilitation published between April 2004 and April 2024.

2.1. Search strategy and selection criteria

To ensure a comprehensive and interdisciplinary review, we employed a combination of controlled vocabulary (MeSH terms) and free-text keywords in our search strategy. Literature was retrieved from PubMed, Cochrane Library, Web of Science, and IEEE Xplore, which collectively cover biomedical sciences, clinical research, engineering, and AI. This selection ensures a balanced representation of both healthcare-focused and technology-driven studies related to AI and gamification in rehabilitation.

To comprehensively capture interdisciplinary studies at the intersection of artificial intelligence and game-based technologies, we adopted a deliberately broad search strategy. Specifically, we used the search string: (game OR gamification) AND (AI OR artificial intelligence). While this approach did not include the keyword “rehabilitation” explicitly, it was chosen to maximize the retrieval of potentially relevant literature across diverse domains such as healthcare, engineering, and human–computer interaction, where relevant studies might not always be indexed under “rehabilitation.” To ensure that the final selection of studies was highly relevant, we applied strict and clearly defined screening criteria during the eligibility assessment phase. Only studies with a primary focus on rehabilitation applications were included. Although this broad strategy initially yielded a large volume of records, it helped reduce the risk of missing pertinent studies that may have used alternative terminologies or were published in interdisciplinary contexts.

Studies were included if they investigated the application of AI and game-based technologies in rehabilitation, focused on rehabilitation treatment as the primary research theme, and were published in English-language peer-reviewed sources. Studies were excluded if they were theses or dissertations (due to the lack of standardized peer review), review articles, or theoretical studies; were unrelated to AI or game-based technologies; did not specifically focus on rehabilitation treatment; were duplicate publications; lacked full-text availability; or had been retracted.

2.2. Data analysis

After retrieval, all references were imported into Zotero, where one researcher conducted an initial screening to remove duplicates. The remaining studies were independently screened by two researchers trained in evidence-based methodology, who assessed eligibility based on the inclusion and exclusion criteria. These researchers also extracted relevant data and evaluated the level of evidence. In cases of disagreement, a third researcher was consulted to reach a final decision. The extracted data included author(s), publication year, title, journal,

research theme, study population, study design, functional category, level of evidence, AI techniques, and gamification approaches. The level of evidence was assessed according to the Oxford Centre for Evidence-Based Medicine classification system [26]. This systematic approach ensured consistency, minimized bias, and maintained the methodological rigor of the review.

3. Results

A total of 4,086 studies were identified through database searches. After removing 603 duplicate records, 3,397 studies were excluded during the initial screening based on titles and abstracts. Following a full-text review, 20 studies met the inclusion criteria and were included in the final analysis [27–46]. The study selection process is illustrated in Fig. 1.

3.1. Overview of publication trends

To further characterize the included studies, we analyzed their publication years, journal categories, countries of first authors, and study design types, as illustrated in Fig. 2(a–d). A summary table (Table 1) has also been provided below Fig. 2 to complement the visual overview with detailed metadata for each study, including author, year, country, journal, and study design.

Most studies ($n = 11$) were published in health-related journals, followed by engineering and technology journals ($n = 6$), suggesting a predominant clinical and technical research orientation (Fig. 2a). The number of publications has increased steadily since 2019, peaking in 2021 (Fig. 2b), which reflects growing academic interest in AI-driven

gamification for rehabilitation. Regarding geographical distribution (Fig. 2c), the majority of first authors were affiliated with institutions in Spain ($n = 6$), China ($n = 3$), and the United States ($n = 2$), reflecting notable contributions from European countries, especially Spain. In terms of study design (Fig. 2d), proof-of-concept studies ($n = 11$) were the most common, followed by clinical trials ($n = 6$) and exploratory studies ($n = 3$), indicating that the field is still in its early stages of empirical development.

3.2. Categorization framework

To ensure a clear and consistent analytical framework, the included studies were categorized into three dimensions: disease or health condition, game-based technology, and AI techniques, as summarized in Table 2. Disease or health condition refers to the primary rehabilitation focus of each study (e.g., stroke, motor dysfunction, cognitive impairment), allowing a structured view of targeted patient groups. Game-based technologies were divided into three types based on their design intent and application: gamification, serious games, and digital games. While some conceptual overlap exists, we distinguished these categories based on the original design purpose and descriptions provided by the study authors. Gamification involves integrating game elements such as points or rewards into non-game contexts to enhance motivation and engagement. Serious games are complete, purpose-built games, which can be digital or non-digital, whose primary purpose is not pure entertainment but rather education, training, or therapy [47]. In contrast, digital games, in the context of our review, refer to commercial or general-purpose games originally created for entertainment that have been repurposed for rehabilitation. AI techniques were grouped into

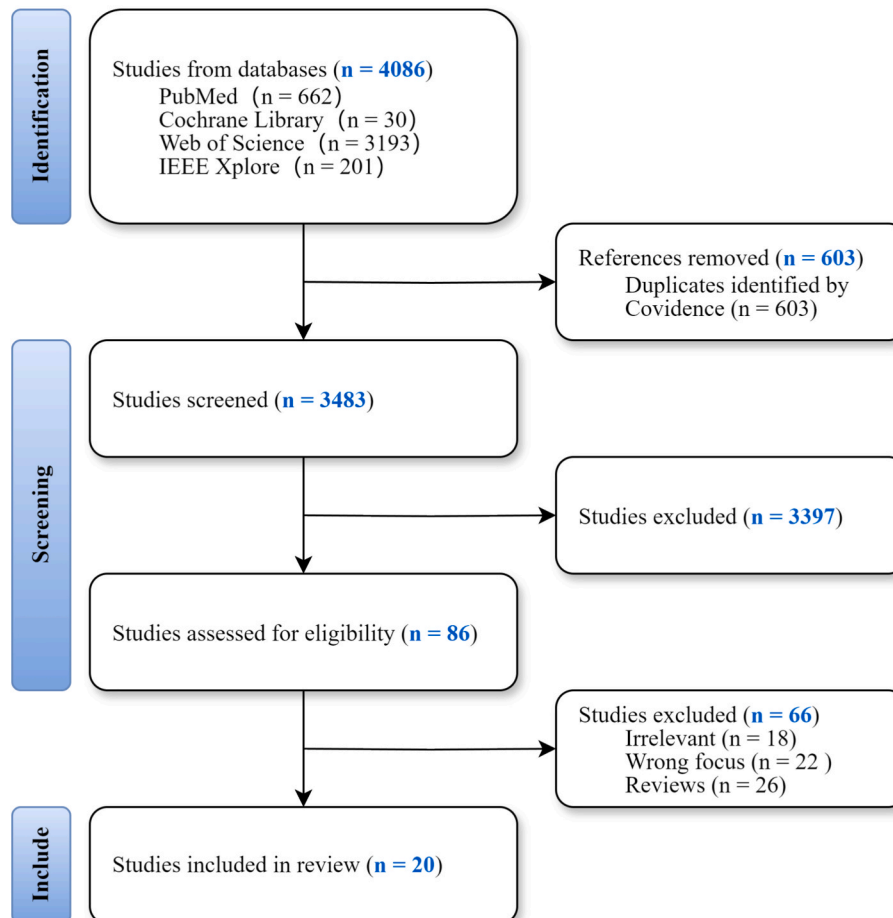


Fig. 1. PRISMA flowchart.

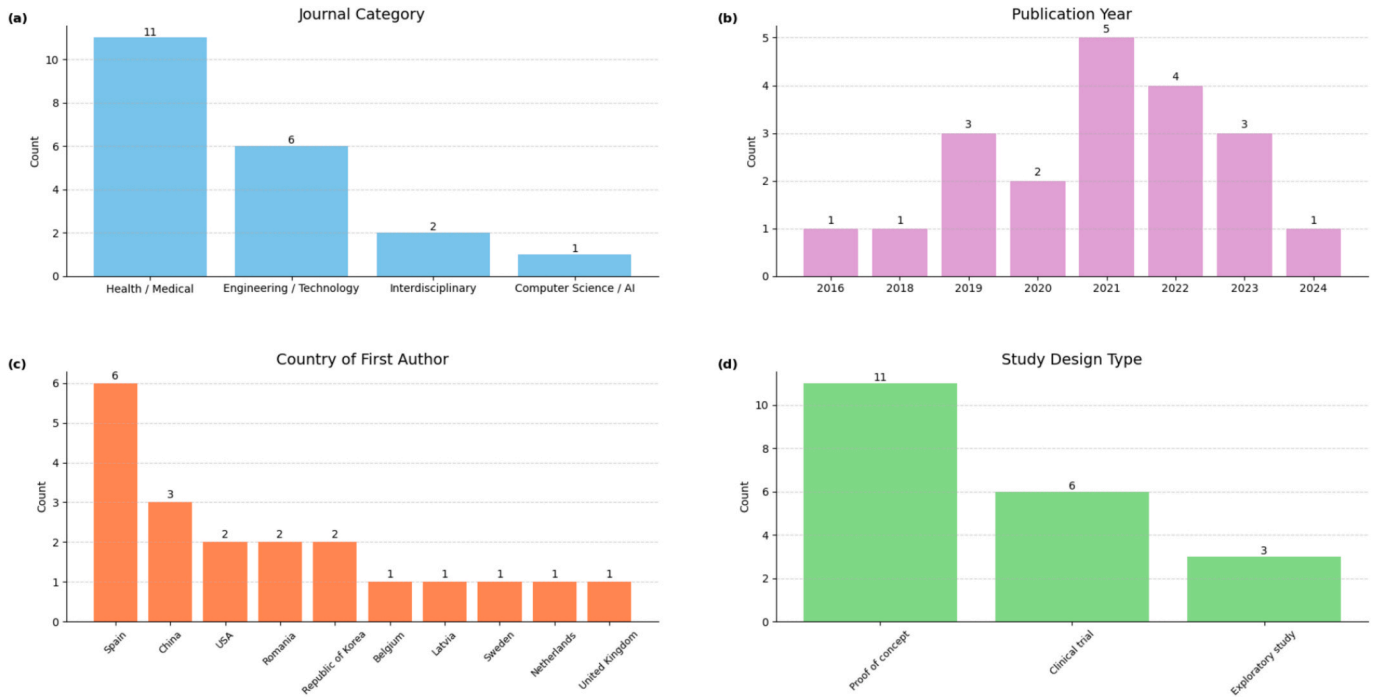


Fig. 2. (a) Journal categories, (b) Publication year, (c) Country of first author, (d) Study design type.

Table 1

Summary of the included studies with details on publication year, country of first author, journal, journal category, and study design.

Study	Year	Country	Journal Name	Journal Category	Study design
Burdea et al. [33]	2021	USA	Assistive Technology	Health/Medical	Proof of concept
Catalán et al. [46]	2023	Spain	Journal of NeuroEngineering and Rehabilitation	Health/Medical	Clinical trial
Chen et al. [29]	2022	China	Frontiers in Physiology	Health/Medical	Proof of concept
De Boi et al. [32]	2024	Belgium	Artificial Intelligence in Medicine	Health/Medical	Proof of concept
Epalte et al. [37]	2023	Latvia	Disability and Rehabilitation: Assistive Technology	Health/Medical	Exploratory study
Eun et al. [34]	2022	Republic of Korea	Journal of Environmental and Public Health	Health/Medical	Proof of concept
Eun et al. [31]	2023	Republic of Korea	Future Generation Computer Systems	Computer Science/AI	Clinical trial
Gómez-Portes et al. [39]	2021	Spain	IEEE Access	Engineering/Technology	Clinical trial
Jung et al. [40]	2019	USA	IEEE Journal of Biomedical and Health Informatics	Health/Medical	Proof of concept
Kristoffersen et al. [43]	2021	Netherlands	Journal of NeuroEngineering and Rehabilitation	Health/Medical	Exploratory study
Lyu et al. [36]	2019	China	Review of Scientific Instruments	Engineering/Technology	Proof of concept
Medina et al. [45]	2021	Spain	Journal of Medical Internet Research	Health/Medical	Clinical trial
Nasri et al. [30]	2020	Spain	SENSORS	Engineering/Technology	Proof of concept
Oliver et al. [28]	2018	Spain	SENSORS	Engineering/Technology	Proof of concept
Ortiz-Catalan et al. [38]	2016	Sweden	LANCET	Interdisciplinary	Clinical trial
Sadeghi Esfahlani et al. [35]	2019	United Kingdom	IEEE Access	Engineering/Technology	Proof of concept
Schez-Sobrinho et al. [27]	2020	Spain	IEEE Access	Engineering/Technology	Exploratory study
Varga et al. [41]	2021	Romania	Public Health and Informatics	Health/Medical	Proof of concept
Varga et al. [42]	2022	Romania	Advances in Informatics, Management and Technology in Healthcare	Health/Medical	Clinical trial
Zhang et al. [44]	2022	China	Advanced Science	Interdisciplinary	Proof of concept

machine learning, deep learning (as a subset of machine learning), and unspecified AI, referring to studies that mentioned AI without providing algorithmic detail. This classification helps clarify the technological scope and practical application of AI-driven gamification in rehabilitation. However, across the included studies, few provided explicit evidence or separate evaluation of how AI components specifically enhanced gamification mechanisms or contributed to improved rehabilitation outcomes. This limitation is further examined in the discussion section.

3.3. Distribution of included studies

The distribution of included studies across different categories is summarized in Table 3. A visual summary of these distributions is provided in Fig. 3, which complements the detailed breakdown shown in Table 2. Regarding disease or health conditions, nine studies [27,29,32,33,36,37,39,40,46] focused on stroke rehabilitation, while two studies [30,44] examined motor dysfunction. Additionally, two studies [41,42] investigated rheumatoid arthritis, and three studies [28,31,34] addressed cognitive impairment. Furthermore, two studies

Table 2
Definitions and descriptions of included literature categories.

Category	Subcategory	Definition
Disease or health condition	Stroke	Ischemic stroke occurs when the blood supply to the brain is interrupted, whereas hemorrhagic stroke results from blood vessel rupture or vascular abnormalities [48].
	Motor dysfunction	A clinical syndrome characterized by excessive or insufficient voluntary and involuntary movements, independent of weakness or spasticity [49].
	Rheumatoid arthritis	A chronic autoimmune disease that primarily affects the joints, often causing joint warmth, swelling, and pain [50].
	Cognitive impairment	A general term describing conditions that affect cognitive processes or various cognitive domains [51].
	Phantom limb pain or limb loss	A phenomenon where patients perceive the presence of a limb despite its amputation or congenital absence [52].
	Attention deficit hyperactivity disorder (ADHD)	A neurodevelopmental disorder characterized by executive dysfunction, manifesting as inattention, hyperactivity, and impulsivity [53].
	Multiple sclerosis	An autoimmune disease that damages the insulating layer of nerve cells in the brain and spinal cord [54].
Game-based technology	Gamification	A strategic approach designed to motivate and engage users by creating experiences similar to gameplay [55].
	Serious games	Games designed primarily with explicit and well-defined educational or training objectives rather than entertainment [56].
	Digital games	Games developed using digital technologies and designed to operate on digital devices [57].
AI techniques	Machine learning	A subfield of AI focused on developing statistical algorithms that enable systems to learn from data and generalize to unseen data [58].
	Deep learning	A subset of machine learning based on neural networks capable of representation learning. The term “deep” refers to the use of multiple-layer structures [59].

Table 3
Categories and distribution proportions of included literature.

Category	% (Number of Studies)
Disease or health condition (n = 20)	
Stroke	45 % (9/20)
Motor dysfunction	10 % (2/20)
Rheumatoid arthritis	10 % (2/20)
Cognitive impairment	15 % (3/20)
Phantom limb pain or limb loss	10 % (2/20)
Attention deficit hyperactivity disorder (ADHD)	5 % (1/20)
Multiple sclerosis	5 % (1/20)
Game-based technology (n = 20)	
Gamification	20 % (4/20)
Serious games	40 % (8/20)
Digital games	40 % (8/20)
AI techniques (n = 20)	
Machine learning	50 % (10/20)
Deep learning	20 % (4/20)
Unspecified AI techniques	30 % (6/20)

[38,43] explored phantom limb pain or limb loss, whereas one study [45] focused on ADHD, and one study [35] investigated multiple sclerosis. In terms of game-based technology, four studies [27,32,37,39] incorporated gamification, while eight studies [28,31,33,34,40–43] utilized serious games. Additionally, eight studies [29,30,35,36,38,44–46] applied digital games. For AI techniques, ten studies [27,28,32,35,38–42,44] employed machine learning, while four studies [29,30,36,43] adopted deep learning. Meanwhile, six studies [31,33,34,45] involved AI techniques without a clearly specified methodology.

To further analyze the distribution of studies, the included literature was categorized and summarized based on disease or health condition, as detailed in Table 4. Additionally, to provide a clearer visual representation, Fig. 4 illustrates the distribution of AI and game-based technologies across different health conditions using a stacked bar chart. This visualization highlights the relative proportions of machine learning, deep learning, and unspecified AI techniques, as well as the use of gamification, serious games, and digital games in each category.

3.4. Insights into stroke rehabilitation

A total of nine studies [27,29,33,36,37,32,39,40,46] explored the role of AI and gamification in stroke rehabilitation, focusing on motor function recovery and cognitive rehabilitation. These studies investigated how AI-driven technologies and game-based interventions can enhance rehabilitation outcomes.

Among them, five studies [27,29,33,36,39] examined motor function recovery, integrating AI to improve training adaptability and effectiveness. For instance, Lyu et al. [36] developed an electromyography (EMG)-controlled knee exoskeleton powered by deep learning, enabling real-time adjustments to training parameters and significantly enhancing lower limb motor function. Similarly, Schez-Sobrinho et al. [27] designed a distributed gamified system utilizing machine learning for remote monitoring of stroke patients' rehabilitation progress, demonstrating strong potential for tele-rehabilitation applications. Meanwhile, four studies [37,32,40,46] focused on cognitive rehabilitation in stroke patients. Jung et al. [40] developed a serious game incorporating machine learning to dynamically adapt its content, aiding patients in cognitive function recovery. Additionally, Catalán et al. [46] conducted a clinical trial assessing the effectiveness of a rehabilitation robot in cognitive function improvement, reporting significant benefits.

Furthermore, the integration of virtual reality (VR) and AI has emerged as a promising approach in stroke rehabilitation. De Boi et al. [32] introduced a gamified system for hemispatial neglect assessment, which proved effective in restoring visuospatial perception. Similarly, Gómez-Portes et al. [39] combined mixed reality with machine learning to develop a remote rehabilitation platform, demonstrating significant improvements in patient outcomes.

3.5. Insights into motor dysfunction rehabilitation

Two studies [30,44] focused on the rehabilitation of lower limb and lumbar motor dysfunction unrelated to stroke. These studies explored the application of AI-driven technologies to enhance rehabilitation outcomes through real-time feedback and personalized training adjustments.

Nasri et al. [30] developed a digital game system controlled via surface electromyography (sEMG), utilizing deep learning for real-time gesture recognition. This system enables patients to perform rehabilitation training using low-cost sensors, significantly improving engagement and motivation. Meanwhile, Zhang et al. [44] introduced a wearable device powered by triboelectric nanogenerator technology, integrating smart insoles to collect gait and lumbar movement data. By leveraging machine learning for data analysis and real-time feedback, this system facilitates lower limb function recovery.

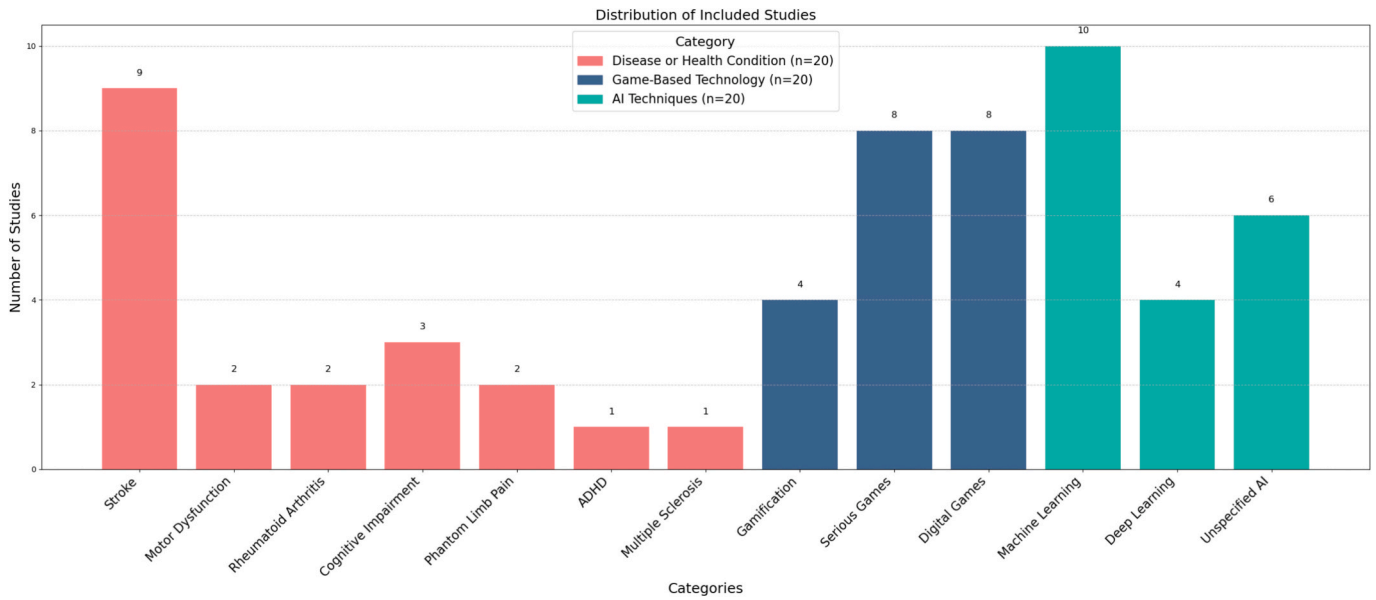


Fig. 3. Distribution of included studies.

3.6. Insights into rheumatoid arthritis rehabilitation

Two studies [41,42] explored the application of AI and serious games in hand function rehabilitation for patients with rheumatoid arthritis (RA). These studies investigated how AI-driven motion analysis and interactive gaming environments can enhance rehabilitation outcomes.

Varga et al. [41] developed a multimodal interactive serious game that utilizes neural networks to classify movements in real-time and provide instant feedback. This system enables patients to perform rehabilitation exercises at home, effectively improving hand flexibility and engagement. Building upon this work, Varga et al. [42] introduced a jump-based serious game designed specifically for RA patients. By leveraging Leap Motion technology to capture hand movements and integrating machine learning to assess motion accuracy, the system effectively enhances grip strength and flexibility. Additionally, the gamified approach boosts motivation and adherence to rehabilitation training.

3.7. Insights into cognitive impairment rehabilitation

Three studies [28,31,34] explored the application of AI and game-based training in cognitive rehabilitation for older adults. These studies examined how AI-driven adaptive training systems can enhance engagement and cognitive function recovery.

Oliver et al. [28] developed an AI-powered home-based serious gaming rehabilitation system, which utilizes real-time analysis and dynamic content adjustment to improve patient engagement and training effectiveness. Expanding on this concept, Eun et al. [34] introduced a gamified cognitive training system, where AI dynamically adapts training content based on individual patient characteristics, effectively enhancing cognitive function. Further advancing this approach, Eun et al. [31] incorporated physical exercise elements into cognitive training and conducted a clinical trial, demonstrating significant improvements in both cognitive and physical function.

3.8. Insights into phantom limb pain and limb loss rehabilitation

Two studies [38,43] explored rehabilitation approaches for phantom limb pain (PLP) patients and upper limb prosthesis users, integrating AI-driven techniques and game-based training to enhance rehabilitation outcomes.

Ortiz-Catalan et al. [38] developed a digital game-based phantom limb movement training system, combining machine learning and augmented reality to treat chronic PLP. The study demonstrated that the system significantly reduced pain intensity, frequency, and interference with daily life, highlighting its potential as a non-invasive therapeutic tool for PLP management. Meanwhile, Kristoffersen et al. [43] introduced a serious game based on EMG pattern recognition, designed to train upper limb prosthesis users in EMG control skills. By employing deep learning for real-time EMG signal analysis and movement feedback, the system enhances patients' muscle control ability, ultimately improving prosthetic proficiency and usability.

3.9. Insights into attention-deficit hyperactivity disorder rehabilitation

Medina et al. [45] conducted a randomized controlled trial (RCT) to investigate the effects of game-based cognitive therapy on children with ADHD. The study utilized a mobile-based digital therapeutic system designed to deliver cognitive training. After 12 weeks of intervention, participants exhibited significant improvements in inhibitory control, cognitive flexibility, and visuospatial working memory.

Furthermore, magnetoencephalography (MEG) findings revealed that enhancements in inhibitory control were associated with increased alpha-band power in the bilateral precuneus and posterior cingulate cortex within the default mode network (DMN). These results suggest that the cognitive training intervention may facilitate brain function recovery through neural modulation.

3.10. Insights into multiple sclerosis rehabilitation

Sadeghi Esfahlani et al. [35] developed a video game-based neuro-rehabilitation system designed to enhance motor and cognitive functions in patients with multiple sclerosis (MS). By integrating AI and machine learning, the system analyzes patient movement data and employs gamification strategies to optimize rehabilitation outcomes. This approach demonstrates the potential of personalized and intelligent therapies in MS rehabilitation.

Table 4

Scoping analysis of included literature, arranged by disease or health topic.

Study	Sample size	Population	Rehabilitation tool	Functional category	Level of evidence	AI technique	Game-based technology	
Stroke (n = 9)								
Schez-Sobrinho et al. [27]; (2020)	27	Stroke patients	Distributed gamified system		Analysis and cognition	3	Machine learning	Gamification
Chen et al. [29]; (2022)	4	Stroke patients	Interactive game with skeletal gesture recognition		Direct intervention	4	Deep learning	Digital games
De Boi et al. [32]; (2024)	38	Stroke patients	Gamified system with virtual reality		Direct intervention	2b	Machine learning	Gamification
Burdea et al. [33]; (2021)	2	Stroke patients	Therapeutic game controller		Direct intervention	4	Unspecified	Serious games
Lyu et al. [36]; (2019)	8	Stroke patients	EMG-controlled knee exoskeleton		Direct intervention	4	Deep learning	Digital games
Epalte et al. [37]; (2023)	12	Stroke patients	Digital assistant		Analysis and cognition	4	Unspecified	Gamification
Catalán et al. [46]; (2023)	14	Stroke patients	Rehabilitation robot		Direct intervention	2b	Unspecified	Digital games
Gómez-Portes et al. [39]; (2021)	25	Stroke patients	Gamified remote rehabilitation system with mixed reality		Direct intervention	4	Machine learning	Gamification
Jung et al. [40]; (2019)	12	Stroke patients	Serious game for cognitive impairment assessment		Analysis and cognition	2b	Machine learning	Serious games
Motor Dysfunction (n = 2)								
Nasri et al. [30]; (2020)	15	Individuals with motor dysfunction	Surface EMG-controlled game		Direct intervention	4	Deep learning	Digital games
Zhang et al. [44]; (2022)	5	Patients with lower limb motor dysfunction	Wearable device using triboelectric nanogenerators		Direct intervention	2	Machine learning	Digital games
Rheumatoid Arthritis (n = 2)								
Varga et al. [41]; (2021)	7	RA patients	Multimodal interactive serious game		Direct intervention	4	Machine learning	Serious games
Varga et al. [42]; (2022)	10	RA patients	Jump-based serious game		Direct intervention	4	Machine learning	Serious games
Cognitive Impairment (n = 3)								
Oliver et al. [28]; (2018)	20	Older adults	Intelligent home-based cognitive rehabilitation system		Direct intervention	4	Machine learning	Serious games
Eun et al. [31]; (2023)	25	Older adults	Cognitive training game		Direct intervention	4	Unspecified	Serious games
Eun et al. [34]; (2022)	37	Older adults	Cognitive training game		Direct intervention	4	Unspecified	Serious games
Phantom Limb Pain or Limb Loss (n = 2)								
Ortiz-Catalan et al. [38]; (2016)	14	Patients with phantom limb pain	Phantom limb motor execution training system		Direct intervention	4	Machine learning	Digital games
Kristoffersen et al. [43]; (2021)	4	Upper limb prosthesis users	Serious game with EMG pattern recognition		Direct intervention	4	Deep learning	Serious games
ADHD (n = 1)								
Medina et al. [45]; (2021)	29	Children with ADHD	Intelligent digital cognitive therapy system		Direct intervention	2	Unspecified	Digital games
Multiple Sclerosis (n = 1)								
Sadeghi Esfahlani et al. [35]; (2019)	52	MS patients	Neurorehabilitation video game		Direct intervention	3	Machine learning	Digital games

4. Discussion

4.1. Addressing RQ1: Current applications and integration of AI-driven gamification in rehabilitation

It is important to interpret these findings with caution, as the current body of evidence is largely composed of preliminary, exploratory studies rather than large-scale clinical trials. Analysis of disease and health conditions in the included studies reveals that the majority of research has focused on stroke rehabilitation [27,29,33,36,37,32,39,40,46], encompassing motor function recovery, cognitive rehabilitation, and other rehabilitation needs. Additionally, studies on motor dysfunction [30,44], RA [41,42], cognitive impairment [28,31,34], phantom limb pain or limb loss [38,43], ADHD [45], and MS [35] have primarily focused on direct intervention strategies.

The findings suggest that AI-driven gamification holds potential to

enhance rehabilitation outcomes through personalized, home-based, and multi-dimensional training programs. Games, as rehabilitation tools, increase patient motivation, provide engaging training experiences, and support multi-faceted therapy approaches [60]. Simultaneously, AI technologies enable real-time monitoring, data-driven analysis, and adaptive training adjustments, delivering personalized feedback while optimizing game mechanics and task difficulty [61]. Furthermore, the integration of multimodal interactive devices with AI-driven gaming systems has been reported to support patient engagement and adherence in early-stage studies. By providing real-time feedback and personalized training experiences, these technologies create a more immersive and interactive rehabilitation environment. Additionally, AI-enabled adaptations in game mechanics and difficulty levels may help improve therapy accessibility and promote adherence, although further clinical validation is still needed.

Despite these promising developments, few of the included studies

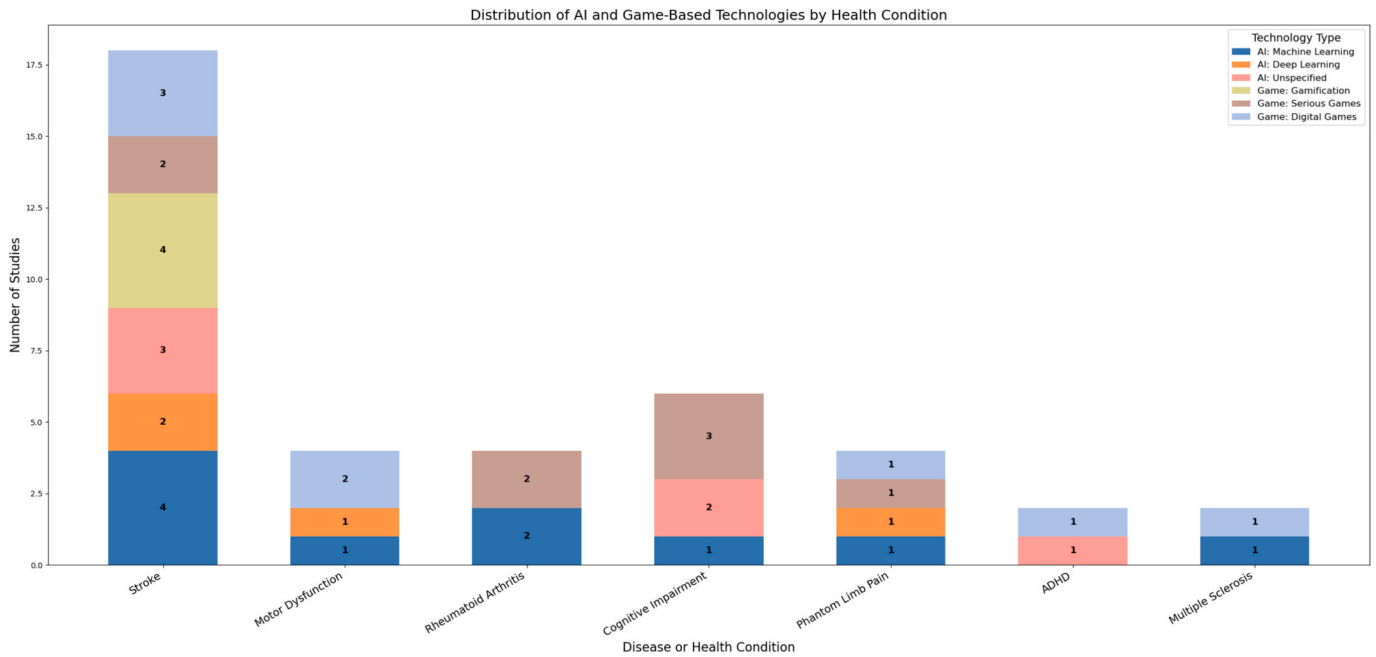


Fig. 4. Distribution of AI and game-based technologies by health condition.

systematically investigate how AI specifically enhances gamification mechanics or contributes to improved rehabilitation outcomes. While some studies [27,36,39] mention features such as real-time feedback, adaptive difficulty adjustment, or performance-based personalization, these components are typically embedded within broader system descriptions and are rarely evaluated in isolation. As such, there remains a lack of empirical evidence demonstrating the additive or synergistic effect of AI within gamified rehabilitation contexts. This observation underscores the gap identified in the Introduction and highlights the need for future research to move beyond technical integration toward mechanism-oriented and outcome-driven evaluation.

4.2. Addressing RQ2: Benefits and challenges of AI-powered gamification in rehabilitation

Existing studies suggest that AI-powered gamification holds several potential advantages in rehabilitation, including enhanced patient engagement, personalized therapy adjustments, and real-time feedback mechanisms [62]. By leveraging machine learning and adaptive algorithms, AI may dynamically adjust training difficulty, support the optimization of rehabilitation exercises, and provide individualized performance insights, which may contribute to greater therapy adherence and improved functional outcomes [63,64]. Moreover, game-based rehabilitation has shown positive potential across various health conditions, including stroke, motor dysfunction, cognitive impairment, and neurodevelopmental disorders [65–67]. Gamified approaches not only increase patient motivation through rewards and interactive challenges but also create multi-sensory learning experiences, which are considered particularly beneficial for motor and cognitive recovery [68,69]. Additionally, remote AI-driven rehabilitation platforms may help improve accessibility for patients with mobility restrictions, allowing home-based therapy while enabling clinicians to monitor progress and adjust interventions as needed [70].

Despite these advantages, several challenges remain in the integration of AI and gamification in rehabilitation. First, most existing studies have small sample sizes and primarily follow proof-of-concept or exploratory designs, with a lack of large-scale randomized controlled trials (RCTs), which limits the generalizability and real-world applicability of findings. Additionally, high heterogeneity across studies,

including variations in disease types, intervention strategies, and participant demographics, poses difficulties in data synthesis and cross-study comparisons, hindering the formation of unified conclusions.

Second, AI applications heavily rely on high-quality datasets, yet in real-world rehabilitation settings, data collection, accuracy, and standardization remain significant challenges [71]. The diverse nature of sensor-based tracking, motion capture systems, and biometric inputs further complicates the development of robust AI models that can generalize across different patient populations [72,72].

Third, the adoption of AI-powered gamification in rehabilitation is constrained by technological barriers and patient-related factors. High implementation costs, hardware requirements, and limited accessibility to advanced equipment create disparities in healthcare availability, particularly in low-resource settings [73]. Moreover, patient acceptance and proficiency with digital rehabilitation platforms vary, with older adults and individuals with cognitive impairments facing usability challenges [74].

Finally, ethical, privacy, and data security concerns require careful consideration when integrating AI with game-based rehabilitation. The collection and processing of sensitive patient data raise ethical dilemmas related to data protection, informed consent, and potential biases in AI decision-making [75]. Furthermore, game-based interventions have received limited discussion regarding ethical implications, particularly in terms of gaming addiction risks, psychological effects, and the long-term impact of immersive technologies in rehabilitation [76,77].

4.3. Addressing RQ3: Research gaps and future directions

Current research on AI-driven gamification in rehabilitation remains in its early stages, with all reviewed studies published within the past eight years and 13 of them appearing in the last three years, reflecting a rapidly growing interest in this field. While promising, the existing literature presents several limitations that warrant further investigation.

One critical gap lies in the lack of clinical trials, which affects the strength and generalizability of current evidence. Among the 20 included studies, only six involved clinical trials, while the majority were proof-of-concept or exploratory in nature. This highlights the preliminary status of research in this area and underscores the need for more rigorous validation. Several factors may contribute to this scarcity,

including difficulties in recruiting appropriate patient populations, ethical constraints related to control group assignment, challenges in ensuring long-term follow-up, and the financial or technical barriers associated with implementing AI-based gamified systems in real-world rehabilitation settings. In addition, the heterogeneity of disease types, outcome measures, and intervention protocols complicates the design and standardization of large-scale trials.

To address these challenges, future research should adopt more robust experimental designs, including randomized controlled trials that directly compare AI-driven gamified interventions with standard rehabilitation protocols or non-gamified digital tools. Outcome measures should encompass both clinical indicators (e.g., functional recovery, cognitive performance) and user-centered metrics (e.g., motivation, adherence, usability), allowing for a more comprehensive evaluation of intervention effectiveness.

Beyond issues of clinical validation, additional technical and design-related challenges must also be addressed. One major challenge is the integration of multimodal data and personalized rehabilitation strategies, which demands the development of more efficient algorithms and user-friendly interfaces [78]. For instance, adaptive game design can enable AI to dynamically assess patient skill levels and adjust training difficulty in real time, ensuring more precise and individualized rehabilitation interventions [7].

Additionally, although AI-enhanced gamification has the potential to expand access to remote and home-based rehabilitation, further advancements are still needed to optimize usability, engagement, and long-term adherence [79]. To establish stronger evidence, future research should include large-scale randomized controlled trials to validate the clinical efficacy of AI-powered gamification and to assess its cost-effectiveness and feasibility in resource-limited settings [80]. As AI and gaming technologies become more integrated into healthcare, it is also essential to address ethical concerns related to data privacy, security, and patient trust [81]. Clear ethical guidelines and regulatory compliance mechanisms will be necessary for broader adoption.

Furthermore, research should continue to optimize personalized rehabilitation strategies, refine remote rehabilitation platforms, and improve multimodal data integration and real-time feedback mechanisms. Advances in wearable technologies, AI-driven motion tracking, and immersive gaming environments hold great potential for enhancing patient engagement and rehabilitation outcomes [79,82,83]. Finally, since most existing studies emphasize short-term outcomes, longitudinal studies are essential to evaluate the sustained effects of AI-driven gamification on rehabilitation performance and adherence over time [84].

4.4. Study limitations and implications

This review has several limitations that warrant consideration.

First, methodological heterogeneity among the included studies presents challenges for synthesis and comparison. Variations in study design, sample characteristics, outcome measures, and intervention modalities limit the generalizability of findings and preclude formal meta-analytic integration. As such, the conclusions of this review should be interpreted as exploratory rather than definitive.

Second, the scope of the search strategy and inclusion criteria may have introduced selection bias. Although four major interdisciplinary databases (PubMed, Cochrane Library, Web of Science, and IEEE Xplore) were searched, relevant studies from other domains such as human-computer interaction, gaming, or sports science may have been overlooked. Furthermore, restricting the review to English-language primary research articles may have excluded pertinent studies published in other languages.

Third, field immaturity and publication lag limit the comprehensiveness of any synthesis in this emerging area. Many AI-driven gamified interventions remain in development or unpublished, and the rapid evolution of AI technologies may outpace academic publication cycles.

Consequently, this review may not fully capture the latest innovations or ongoing projects.

Despite these limitations, this scoping review offers a timely synthesis of current applications, identifies critical gaps, and suggests directions for future research. Future reviews would benefit from broader search strategies, the inclusion of gray literature, and the adoption of more standardized reporting and evaluation frameworks.

5. Conclusions

This scoping review systematically examined the applications, benefits, challenges, and future directions of AI-driven gamification in rehabilitation. These technologies have been increasingly adopted across diverse domains—including stroke, motor dysfunction, rheumatoid arthritis, cognitive impairment, phantom limb pain, ADHD, and multiple sclerosis—offering advantages such as enhanced patient engagement, improved therapy adherence, along with the prospect of more effective outcomes enabled by personalized interventions, real-time feedback, and adaptive difficulty adjustment.

Nonetheless, several barriers hinder widespread implementation. Small sample sizes, methodological heterogeneity, and dependence on high-quality data limit the generalizability of current findings and complicate large-scale clinical translation. Additionally, concerns surrounding data privacy, ethical standards, technological access, and cost-effectiveness must be addressed to support equitable and sustainable adoption. The lack of large-scale randomized controlled trials further restricts the ability to draw definitive conclusions about long-term effectiveness.

Crucially, these promising findings must be viewed in the context of the available evidence. As this review highlights, the field is dominated by proof-of-concept and exploratory studies (Level 4 evidence), meaning that claims of significant clinical advancement are premature. The lack of large-scale randomized controlled trials remains a major barrier to drawing definitive conclusions about long-term effectiveness.

To advance the field, future research should prioritize rigorous clinical trials, enhanced multimodal data integration, and the development of AI-driven remote rehabilitation platforms. Emerging technologies such as wearable sensors, motion tracking, and immersive gaming hold strong potential for scalable, accessible, and personalized rehabilitation. As these tools evolve, interdisciplinary collaboration among healthcare professionals, engineers, and game designers will be essential for shaping next-generation intelligent rehabilitation solutions.

This review provides the first structured synthesis of AI-driven gamification in rehabilitation, integrating disease domains, AI techniques, and game-based modalities. Through systematic categorization and visual mapping, it offers a comprehensive overview of the current landscape and identifies key underexplored intersections. By highlighting the lack of empirical validation for AI-enhanced gamification mechanisms, this review lays the groundwork for future outcome-driven, mechanism-oriented research in the field.

CRediT authorship contribution statement

Minkai Wang: Writing – original draft, Investigation, Formal analysis, Data curation. **Jingdong Zhu:** Writing – review & editing, Methodology, Data curation. **Wei Qian:** Writing – review & editing, Methodology, Data curation. **Hanjie Gu:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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